भारतीय प्रौद्योगिकी संस्थान जोधपुर
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Sukesh P (M24IM1007) <m24im1007@iitj.ac.in>

nam spectra

6 messages

Sukesh P (M24IM1007) <m24im1007@iitj.ac.in> Wed, Apr 15, 2026 at 6:18 PM
To: Sukesh Periyasamy <sukeshperiyasamy@gmail.com>

https://pmc.ncbi.nlm.nih.gov/articles/PMC12903052/

Sukesh P (M24IM1007) <m24im1007@iitj.ac.in> Thu, Apr 16, 2026 at 1:11 AM
To: Sukesh Periyasamy <sukeshperiyasamy@gmail.com>

https://www.sciencedirect.com/science/article/pii/S0924203124000237?via%3Dihub

On Wed, Apr 15, 2026 at 6:18 PM Sukesh P (M24IM1007) <m24im1007@iitj.ac.in> wrote:

> https://pmc.ncbi.nlm.nih.gov/articles/PMC12903052/

Sukesh P (M24IM1007) <m24im1007@iitj.ac.in> Thu, Apr 16, 2026 at 1:15 AM
To: Sukesh Periyasamy <sukeshperiyasamy@gmail.com>

https://www.sciencedirect.com/science/article/pii/S0924203124000237?via%3Dihub
this one has dft simulated

[Quoted text hidden]

Sukesh P (M24IM1007) <m24im1007@iitj.ac.in> Thu, Apr 16, 2026 at 10:58 AM
To: Sukesh Periyasamy <sukeshperiyasamy@gmail.com>

https://link.springer.com/article/10.1007/s10103-020-03028-9
NAM is a major component of bacterial peptidoglycan, so its Raman signature appears in these peaks
[Quoted text hidden]

Sukesh P (M24IM1007) <m24im1007@iitj.ac.in> Thu, Apr 16, 2026 at 11:04 AM
To: Sukesh Periyasamy <sukeshperiyasamy@gmail.com>

Peak (cm^{-1})	Assignment
540	Glycosidic bond (NAM–NAG)
1380	Peptidoglycan (NAM structure)

🟡 OPTIONAL (VERY CAREFUL USE)

You may **mention but not claim as NAM**:

Peak	Use
1129	General carbohydrate/lipid
1239	Amide (support only)

[Quoted text hidden]

Sukesh P (M24IM1007) <m24im1007@iitj.ac.in> Thu, Apr 16, 2026 at 11:08 AM
To: Sukesh Periyasamy <sukeshperiyasamy@gmail.com>

https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0005470

✅ Common Peaks (Core fingerprint peaks)

These peaks appear consistently across bacteria:

626, 655, 960, 1026, 1240, 1330, 1396, 1460

👉 These are the **main fingerprint peaks** for bacterial identification

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✅ Gram-negative Specific Peaks

631, 655, 725, 960, 1095, 1130, 1330, 1460

👉 Important for **E. coli / Gram-negative analysis**

✅ Dynamic / Growth-dependent Peaks

(used to monitor bacterial growth changes)

655, 1130, 1219, 1245 (increase)
725, 1095 (decrease)

✅ Antibiotic Response Peaks

(very useful for diagnostics)

725, 1095 (decrease during inhibition)
732 (reference / normalization peak)

⭐ Special Peak (Very Important)

~730–732 cm^{-1}

- Used as **normalization reference**
- Related to **cell wall (peptidoglycan / adenine)**
- Strong biomarker for bacterial detection

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